STATISTICAL LEARNING IN MARKETING

ASSIGNMENT 3

S6560245

MANAGERIAL QUESTION:
HOW DO OUR OWN AND OUR COMPETITORS' ADVERTISING ACTIVITIES DYNAMICALLY INFLUENCE OUR SALES OVER A QUARTER?

METHODOLOGICAL APPROACH

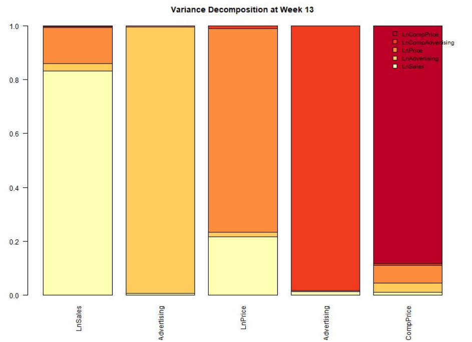
There are five important (log-transformed) key variables in the dataset: Ixmør's sales, advertising, and price, along with competitor advertising and competitor price. To identify the primary drivers of Ixmør's sales over a 13-week horizon, the following methodological steps were taken:

1. Granger causality tests were conducted to determine which variables are temporally causing which other variables. All five key variables are treated as endogenous to account for complex, simultaneous feedback loops within the market system.
2. Before estimating the dynamic system, it is checked if all series are stationary and have no unit roots. To ensure robust results and determine if variables are stationary or evolving, a testing approach was used including Augmented Dickey-Fuller (ADF), Phillips-Perron (PP), and KPSS tests.
3. A Vector Autoregression (VAR) model was then estimated, which includes control variables to account for external influences like seasonality. The optimal lag length was determined using the Bayesian Information Criterion (BIC), which performs well for both order selection and forecasting. The BIC indicated that a lag of one was the optimal choice for this specific model.
4. Impulse Response Functions (IRFs) were computed to focus on the net effect over time of a change in one endogenous variable through the whole system. Because the variables are stationary, the effect dies out over time, making the cumulative effect the most relevant metric to analyze. Since IRFs contain lots of uncertainty in the parameter estimates, a 68% significance level was used, as 95% may be too harsh.
5. Finally, Forecast Error Variance Decomposition (FEVD) was conducted to show the relative importance of all the endogenous variables in contributing to the variation in Ixmør's sales. This quantified what part of the forecast error variance is due to own past shocks versus all past shocks to the other endogenous variables at the end of the 13-week quarter.

CONCLUSION OF THE ANALYSIS

- **Granger causality** shows that while advertising drives sales only for the first five weeks before its effect completely decays, pricing maintains an immediate and highly significant impact across the entire quarter. This persistent relationship proves that consumers have a long-lasting 'price memory'.
- The **VAR model** explains about 46% of weekly sales variance. Own price has a strong and highly significant negative effect on sales, indicating high price sensitivity. Past sales positively influence current sales, reflecting consumer loyalty. Advertising shows only a weak negative effect, while competitor price and advertising have no significant impact.
- **IRF analysis** reveals our market is highly price-sensitive but entirely immune to advertising and competitor actions. While price increases cause a severe two-week sales drop and past sales generate positive short-term momentum, shocks to ad spend or rival tactics yield absolutely no significant volume impact.
- **FEVD** analysis confirms that sales variance is driven almost entirely by market inertia and internal pricing strategy (see the graph). Advertising and competitor actions show no significant impact, proving that our performance is dictated by historical momentum and price sensitivity rather than short-term tactical spend or external rival pressure.

The brand manager is <u>**NOT**</u> correct in the belief that the most important driver, both by Ixmør itself and by competitors, is advertising.
The analysis clearly shows that the main driver of Ixmør's sales is its own past sales, while pricing strategies play a secondary role.



MANAGERIAL ADVICE

Managerial implications

Since the IRF and FEVD analyses show advertising has absolutely no significant volume impact and Granger causality limits its effect to just five weeks, continuous heavy spending on tactical ads is a waste of capital.

Ixmør should:

- **Prioritize pricing strategy**: Consumers are highly price-sensitive, pricing decisions should be the main strategic lever, avoiding aggressive price increases that could significantly reduce sales. (Gauri, D. K., et al. 2017).
- **Use temporary promotions instead of permanent price changes**: Short-term discounts may stimulate demand while preventing the long-term sales losses associated with sustained price increases. (Lamey, L., et al. 2018).
- **Strengthen customer retention**: Because past sales strongly influence current sales, maintaining customer loyalty and repeat purchasing should be a key strategic objective. (Datta, H., Ailawadi, K. L., & van Heerde, H. J. 2017).
- **Re-evaluate advertising investments**: Advertising shows only weak and short-lived effects, suggesting that current spending may not be efficient and should be reassessed or reallocated. (Shapiro, B. T., Hitsch, G. J., & Tuchman, A. E. 2021)
- **Focus less on competitor actions**: Competitor pricing and advertising have no significant impact on sales, managerial decisions should concentrate primarily on internal strategies rather than reacting to competitors. (Steenkamp, J. B. E., & Slotegraaf, R. J. 2019).

USE OF AI

Artificial Intelligence (ChatGPT and Gemini) was used as supportive tools during this assignment. All analyses, coding, and final interpretations were made independently by me. AI was only used to check and improve my work. More specifically, AI was used for the following tasks:

Help to define the managerial question

- **Pro:** Helped to formulate a clear and testable managerial question based on the statement of the brand manager of Ixmør
- **Con:** Risk of framing the question differently than originally intended. Therefore, I critically evaluated and adjusted the AI suggestions to align with the core business problem.

Double-checking the R coding

- **Pro:** Significantly sped up the debugging process when encountering errors in the code and it helped to understand needed libraries.
- **Con:** AI often suggested inaccurate programming techniques or complex functions that were not discussed in the lectures.

Helped with the interpretation of the results

- **Pro:** Assisted in interpreting the coefficients of the VAR model and the dynamics of the IRF's
- **Con:** Risk of misinterpreting results. Therefore, all interpretations were carefully checked with the lecture content.

Helped with suggestions for the manager

- **Pro:** AI provided initial strategic suggestions for Ixmør based on the findings.
- **Con:** Suggestions lacked specific industry creativity. Therefore, I refined the advice to be more actionable and grounded on data findings.

Literature

- **Pro:** AI assisted in summarizing and extracting key points from the academic articles found.
- **Con:** AI is not always reliable for finding or verifying literature support. Therefore, I independently identified and verified all academic sources to ensure their relevance.

01 APPENDIX

ALL R CODE

R CODE: IMPORTING DATA & STEP 1

```
#packages
library(vars)
library(lmtest)
library(aTSA)
library(gplots)
library(RColorBrewer)

# Load data
ts_data <- read.csv(file.choose(), header = TRUE)

# Data overview
summary(ts_data)

# Plot main variables over time
plot(ts_data$Week, ts_data$LnSales, type="l", col="blue",
lwd=5, xlab="weeks", ylab="LnSales", main="Sales over time")
plot(ts_data$Week, ts_data$LnAdvertising, type="l", col="red",
lwd=5, xlab="weeks", ylab="LnAdvertising", main="Advertising
over time")
plot(ts_data$Week, ts_data$LnPrice, type="l", col="darkgreen",
lwd=5, xlab="weeks", ylab="LnPrice", main="Price over time")
```

```
# Granger causality: Adv -> Sales
grangmat.df        <- data.frame(matrix(0,nrow = 13, ncol=2))
names(grangmat.df) <- c("nlags","pval")
for (nlags in 1:13){
  grangout              <- grangertest(LnSales ~ LnAdvertising, order =
nlags, data = ts_data)
  grangmat.df[nlags,1]  <- nlags
  grangmat.df[nlags,2]  <- grangout[2,4]
  print(grangout)
}

# Granger causality: Price -> Sales
grangmat.df        <- data.frame(matrix(0,nrow = 13, ncol=2))
names(grangmat.df) <- c("nlags","pval")
for (nlags in 1:13){
  grangout              <- grangertest(LnSales ~ LnPrice, order = nlags,
data = ts_data)
  grangmat.df[nlags,1]  <- nlags
  grangmat.df[nlags,2]  <- grangout[2,4]
  print(grangout)
}
```

R CODE: STEP 2

```
# Unit root tests (ADF, PP, KPSS)

# Sales
adf.test(ts_data$LnSales, output = TRUE)
pp.test(ts_data$LnSales, output = TRUE)
kpss.test(ts_data$LnSales, output = TRUE)

# Advertising
adf.test(ts_data$LnAdvertising, output = TRUE)
pp.test(ts_data$LnAdvertising, output = TRUE)
kpss.test(ts_data$LnAdvertising, output = TRUE)

# Price
adf.test(ts_data$LnPrice, output = TRUE)
pp.test(ts_data$LnPrice, output = TRUE)
kpss.test(ts_data$LnPrice, output = TRUE)

# Competitor Advertising
adf.test(ts_data$LnCompAdvertising, output = TRUE)
pp.test(ts_data$LnCompAdvertising, output = TRUE)
kpss.test(ts_data$LnCompAdvertising, output = TRUE)

# Competitor Price
adf.test(ts_data$LnCompPrice, output = TRUE)
pp.test(ts_data$LnCompPrice, output = TRUE)
kpss.test(ts_data$LnCompPrice, output = TRUE)
```

R CODE: STEP 3

```
# VAR Model Setup

# Define variables
endo_vars = ts_data[, c("LnSales", "LnAdvertising", "LnPrice", "LnCompAdvertising", "LnCompPrice")]
exo_vars = as.matrix(ts_data[, c("Qrtr1", "Qrtr2", "Qrtr3")])

# Select optimal lags
lag_selection <- VARselect(endo_vars, lag.max = 13, type = "both", exogen = exo_vars)
print(lag_selection$selection)

# Estimate VAR model
var_model <- VAR(endo_vars, p = 1, type = "both", exogen = exo_vars)
summary(var_model, "LnSales")
```

R CODE: STEP 4

```
# Cumulative IRF
# 1. Shock from own Sales
irf_sales_cum <- irf(var_model, impulse = "LnSales", response =
"LnSales", n.ahead = 13,
                     ortho = TRUE, cumulative = TRUE, boot = TRUE,
ci = 0.68, runs = 500)
plot(irf_sales_cum, main = "Cumulative impact of Own Sales on
Sales")
# 2. Shock from Advertising
irf_adv_cum <- irf(var_model, impulse = "LnAdvertising",
response = "LnSales", n.ahead = 13,
                   ortho = TRUE, cumulative = TRUE, boot = TRUE, ci
= 0.68, runs = 500)
plot(irf_adv_cum, main = "Cumulative impact of Advertising on
Sales")
# 3. Shock from Price
irf_price_cum <- irf(var_model, impulse = "LnPrice", response =
"LnSales", n.ahead = 13,
                     ortho = TRUE, cumulative = TRUE, boot = TRUE,
ci = 0.68, runs = 500)
plot(irf_price_cum, main = "Cumulative impact of Price on
Sales")
```

```
# 4. Shock from Comp. Advertising
irf_compadv_cum <- irf(var_model, impulse =
"LnCompAdvertising", response = "LnSales", n.ahead = 13,
                       ortho = TRUE, cumulative = TRUE, boot = TRUE,
ci = 0.68, runs = 500)
plot(irf_compadv_cum, main = "Cumulative impact of Comp.
Adv. on Sales")

# 5. Shock from Comp. Price
irf_compprice_cum <- irf(var_model, impulse = "LnCompPrice",
response = "LnSales", n.ahead = 13,
                         ortho = TRUE, cumulative = TRUE, boot =
TRUE, ci = 0.68, runs = 500)
plot(irf_compprice_cum, main = "Cumulative impact of Comp.
Price on Sales")
```

R CODE: STEP 5

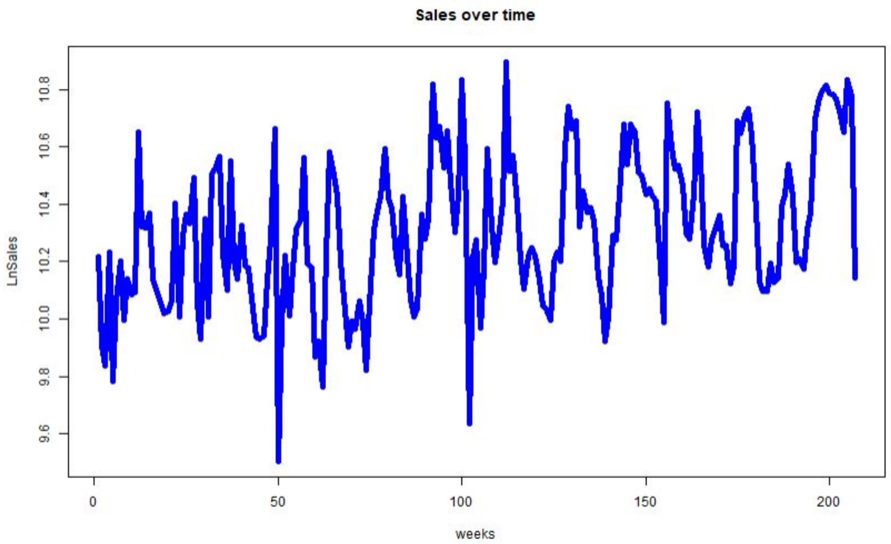
```
# FEVD (Forecast Error Variance Decomposition)

fevd_sales <- fevd(var_model, n.ahead = 13)

# Print weeks 1, 6, and 13 for all variables to check system dynamics
lapply(fevd_sales, function(x) x[c(1, 6, 13), ])

# Extract week 13 data for the final stacked barplot
fevd_week13 <- sapply(fevd_sales, function(x) x[13, ])

# Stacked barplot at week 13
barplot(fevd_week13,
        col = brewer.pal(5, "YlOrRd"),
        main = "Variance Decomposition at Week 13",
        las = 2,
        legend.text = rownames(fevd_week13),
        args.legend = list(x = "topright", bty = "n", cex = 0.8))
```

02 APPENDIX

ALL RESULTS AND TABLES/FIGURES

RESULTS: SALES OVER TIME

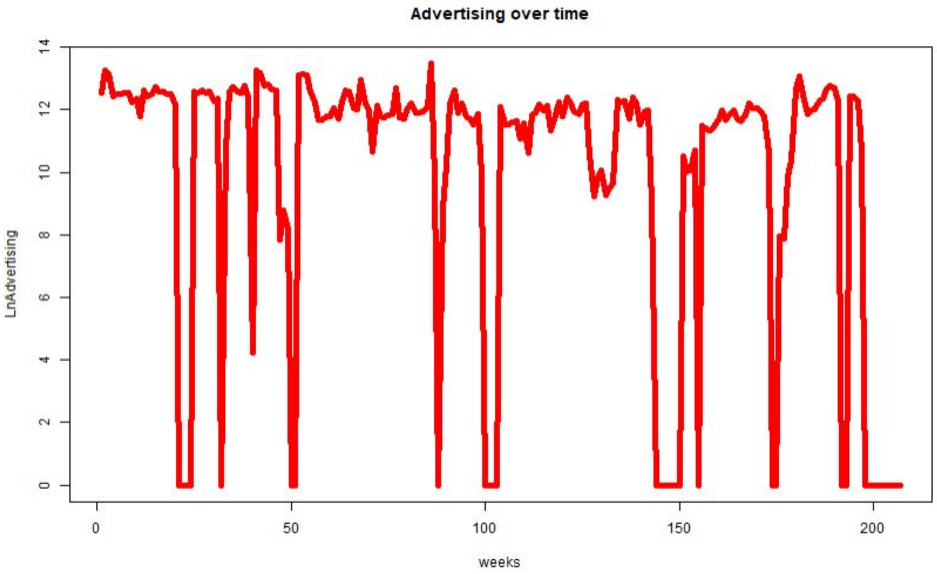
The time series plots of log-transformed variables over 207 weeks reveal the complex dynamic system of Ixmør's market performance.



- The Sales over time plot exhibits significant fluctuations with periodic peaks and troughs.
- Particularly sharp declines are observed around weeks 50, 100, and 150, suggesting the influence of seasonality or external market influences on consumer behavior.
- While sales show notable volatility, an overall upward trend is observed towards the end of the period, where sales stabilize at a higher level.
- The Advertising over time plot shows a highly intermittent and "spiky" pattern.
- This indicates that marketing efforts are concentrated in specific promotional bursts rather than being a continuous effort throughout the year.
- The Price over time plot displays a general downward trend with high volatility.
- Sharp drops around week 100 and week 150 often coincide with shifts in sales, suggesting that price adjustments are used as a reactive lever to stimulate demand during downturns.
- Given these dynamics, Ixmør should investigate the specific drivers of sales performance and assess the long-term effectiveness of its intermittent advertising bursts to ensure stability in the organic food sector.

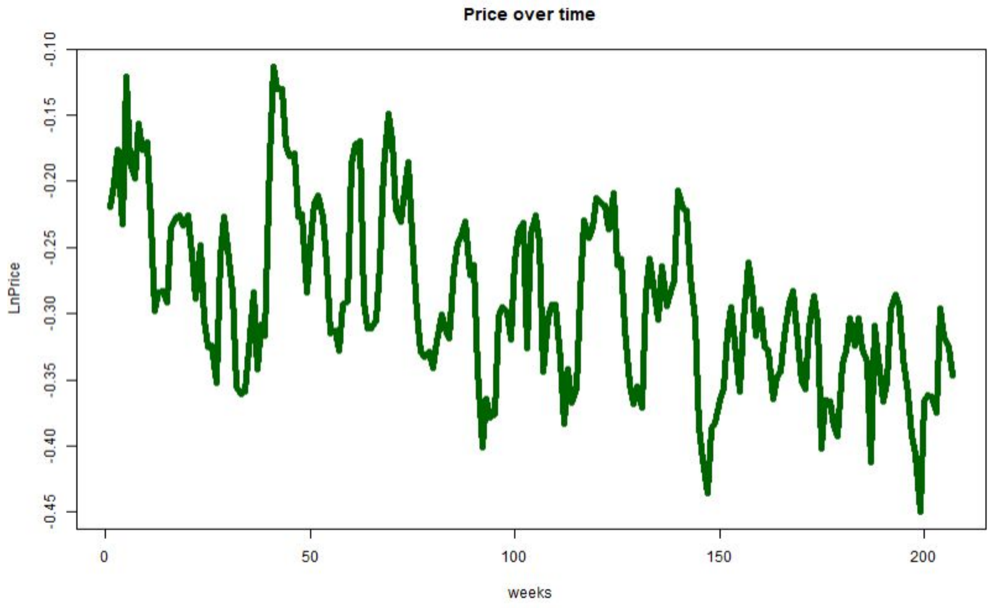
RESULTS: ADVERTISING OVER TIME

- Instead of continuous spending, the data shows sharp, periodic spikes followed immediately by sudden drops to zero, indicating frequent periods of complete media inactivity.
- This recurring "on-and-off" behavior suggests that marketing efforts are concentrated into intense promotional bursts, characteristic of a "flighting" or pulsing media strategy.
- While the maximum intensity of the spikes remains relatively consistent throughout the four-year period, the extreme volatility points to a highly tactical, rather than "always-on", campaign execution.
- Given these dynamics, Ixmør should investigate the specific return on investment of these concentrated bursts and assess the long-term effectiveness of its intermittent advertising to ensure optimal resource allocation in the organic food sector.



RESULTS: PRICE OVER TIME

- While the baseline price appears to remain relatively stable at its upper limit, the data shows recurrent, sharp downward spikes, indicating periods of aggressive discounting or promotional pricing.
- These sudden drops in price suggest a highly active and tactical pricing strategy, likely deployed as a reactive lever to temporarily stimulate consumer demand or respond to market pressures.
- Throughout the entire period, the frequency and depth of these price cuts continue without signs of stabilization, highlighting a sustained reliance on price-based incentives to drive volume.
- Given these dynamics, Ixmør should investigate the specific drivers of this extreme pricing volatility, assess the long-term profitability and brand impact of frequent deep discounts, and evaluate strategies to build brand value that is less dependent on continuous price promotions in the organic food sector.



- Crucially, a comparative visual analysis seems to reveal a strong inverse relationship between extreme price cuts and sales volume. The deepest downward spikes in price, specifically observed around weeks 50, 100, and 150, coincide with the most significant upward spikes in the sales plot

RESULTS: GRANGER CAUSALITY ADV <- SALES

```
Granger causality test
Model 1: LnSales ~ Lags(LnSales, 1:1) +
Lags(LnAdvertising, 1:1)
Model 2: LnSales ~ Lags(LnSales, 1:1)
  Res.Df Df       F    Pr(>F)
1    203
2    204 -1 9.5372 0.002295 **
```

```
Model 1: LnSales ~ Lags(LnSales, 1:2) +
Lags(LnAdvertising, 1:2)
Model 2: LnSales ~ Lags(LnSales, 1:2)
  Res.Df Df       F    Pr(>F)
1    200
2    202 -2 5.3316 0.005548 **
```

```
Model 1: LnSales ~ Lags(LnSales, 1:3) +
Lags(LnAdvertising, 1:3)
Model 2: LnSales ~ Lags(LnSales, 1:3)
  Res.Df Df       F   Pr(>F)
1    197
2    200 -3 3.6474 0.01361 *
```

```
Model 1: LnSales ~ Lags(LnSales, 1:4) +
Lags(LnAdvertising, 1:4)
Model 2: LnSales ~ Lags(LnSales, 1:4)
  Res.Df Df       F   Pr(>F)
1    194
2    198 -4 2.9787 0.02039 *
```

```
Model 1: LnSales ~ Lags(LnSales, 1:5) +
Lags(LnAdvertising, 1:5)
Model 2: LnSales ~ Lags(LnSales, 1:5)
  Res.Df Df       F   Pr(>F)
1    191
2    196 -5 2.3908 0.03936 *
```

```
Model 1: LnSales ~ Lags(LnSales, 1:6) +
Lags(LnAdvertising, 1:6)
Model 2: LnSales ~ Lags(LnSales, 1:6)
  Res.Df Df       F   Pr(>F)
1    188
2    194 -6 2.0519 0.06085 .
```

During the first five weeks, advertising displays a significant causal link to sales, after that, the p-values become marginal.

RESULTS: GRANGER CAUSALITY: PRICE -> SALES

```
Granger causality test
Model 1: LnSales ~ Lags(LnSales, 1:1) +
Lags(LnPrice, 1:1)
Model 2: LnSales ~ Lags(LnSales, 1:1)
  Res.Df Df       F     Pr(>F)
1    203
2    204 -1 23.314 2.704e-06 ***
```

```
Model 1: LnSales ~ Lags(LnSales, 1:2) +
Lags(LnPrice, 1:2)
Model 2: LnSales ~ Lags(LnSales, 1:2)
  Res.Df Df       F     Pr(>F)
1    200
2    202 -2 11.232 2.383e-05 ***
```

```
Model 1: LnSales ~ Lags(LnSales, 1:3) +
Lags(LnPrice, 1:3)
Model 2: LnSales ~ Lags(LnSales, 1:3)
  Res.Df Df      F     Pr(>F)
1    197
2    200 -3 7.282 0.0001175 ***
```

```
Model 1: LnSales ~ Lags(LnSales, 1:4) +
Lags(LnPrice, 1:4)
Model 2: LnSales ~ Lags(LnSales, 1:4)
  Res.Df Df      F     Pr(>F)
1    194
2    198 -4 5.6956 0.0002352 ***
```

```
Model 1: LnSales ~ Lags(LnSales, 1:5) +
Lags(LnPrice, 1:5)
Model 2: LnSales ~ Lags(LnSales, 1:5)
  Res.Df Df     F     Pr(>F)
1    191
2    196 -5 4.975 0.0002632 ***
```

```
Model 1: LnSales ~ Lags(LnSales, 1:6) +
Lags(LnPrice, 1:6)
Model 2: LnSales ~ Lags(LnSales, 1:6)
  Res.Df Df      F     Pr(>F)
1    188
2    194 -6 4.2373 0.0005015 ***
```

RESULTS: GRANGER CAUSALITY: PRICE -> SALES

```
Model 1: LnSales ~ Lags(LnSales, 1:7) +
Lags(LnPrice, 1:7)
Model 2: LnSales ~ Lags(LnSales, 1:7)
  Res.Df Df      F   Pr(>F)
1    185
2    192 -7 3.3872 0.001995 **
```

```
Model 1: LnSales ~ Lags(LnSales, 1:8) +
Lags(LnPrice, 1:8)
Model 2: LnSales ~ Lags(LnSales, 1:8)
  Res.Df Df      F   Pr(>F)
1    182
2    190 -8 2.9653 0.003828 **
```

```
Model 1: LnSales ~ Lags(LnSales, 1:9) +
Lags(LnPrice, 1:9)
Model 2: LnSales ~ Lags(LnSales, 1:9)
  Res.Df Df     F   Pr(>F)
1    179
2    188 -9 2.937 0.002833 **
```

```
Model 1: LnSales ~ Lags(LnSales, 1:10) +
Lags(LnPrice, 1:10)
Model 2: LnSales ~ Lags(LnSales, 1:10)
  Res.Df  Df     F   Pr(>F)
1    176
2    186 -10 2.616 0.005472 **
```

```
Model 1: LnSales ~ Lags(LnSales, 1:11) +
Lags(LnPrice, 1:11)
Model 2: LnSales ~ Lags(LnSales, 1:11)
  Res.Df  Df     F  Pr(>F)
1    173
2    184 -11 2.133 0.02034 *
```

```
Model 1: LnSales ~ Lags(LnSales, 1:12) +
Lags(LnPrice, 1:12)
Model 2: LnSales ~ Lags(LnSales, 1:12)
  Res.Df  Df      F  Pr(>F)
1    170
2    182 -12 2.2103 0.01321 *
```

RESULTS: GRANGER CAUSALITY: PRICE -> SALES

```
Model 1: LnSales ~ Lags(LnSales, 1:13) +
Lags(LnPrice, 1:13)
Model 2: LnSales ~ Lags(LnSales, 1:13)
  Res.Df  Df      F  Pr(>F)
1    167
2    180 -13 2.1895 0.01194 *
```

- The Granger causality test output over the 13-week lag period reveals a highly robust and persistent relationship where pricing temporally causes sales.
- Unlike the advertising effect, which completely decays after five weeks, price displays a strictly significant causal link to sales performance across all 13 tested lags.
- The predictive power is strong in the immediate short term (weeks 1 through 6), with extreme p-values nearing zero ($p < 0.001$), confirming that consumers react instantly and aggressively to price changes.
- Furthermore, this temporal effect maintains its statistical significance throughout the entire quarter, indicating a long-lasting consumer memory for pricing adjustments.
- Given these dynamics, Ixmør should recognize price as its most powerful and persistent marketing lever, evaluating the long-term volume impact of every pricing decision and avoiding excessive discounting that could permanently alter consumer price expectations in the organic food sector.

RESULTS: UNIT ROOT TESTS <- SALES

```
Augmented Dickey-Fuller Test
 Type 1: no drift no trend
      lag      ADF p.value
[1,]    0 -0.1863   0.590
[2,]    1 -0.0186   0.638
[3,]    2  0.1166   0.677
[4,]    3  0.0172   0.649
[5,]    4  0.1907   0.698
Type 2: with drift no trend
      lag    ADF p.value
[1,]    0 -7.21     0.01
[2,]    1 -6.23     0.01
[3,]    2 -5.32     0.01
[4,]    3 -5.66     0.01
[5,]    4 -5.29     0.01
Type 3: with drift and trend
      lag    ADF p.value
[1,]    0 -8.12     0.01
[2,]    1 -7.09     0.01
[3,]    2 -6.11     0.01
[4,]    3 -6.79     0.01
[5,]    4 -6.37     0.01
```

```
Phillips-Perron Unit Root Test
alternative: stationary

Type 1: no drift no trend
 lag   Z_rho p.value
   4 -0.033    0.683
-----
 Type 2: with drift no trend
 lag Z_rho p.value
   4 -85.3     0.01
-----
 Type 3: with drift and trend
 lag Z_rho p.value
   4   -105    0.01
```

```
KPSS Unit Root Test
alternative: nonstationary

Type 1: no drift no trend
 lag    stat p.value
   3 0.0485      0.1
-----
 Type 2: with drift no trend
 lag   stat p.value
   3 0.874     0.01
-----
 Type 1: with drift and trend
 lag    stat p.value
   3 0.0249      0.1
```

- ADF & PP Tests: Both strongly reject the unit root hypothesis ($p \leq 0.01$ with drift and trend).
- KPSS Test: Further validates these findings, failing to reject and thus confirming stationarity ($p \geq 0.10$).
- LnSales is stationary.

RESULTS: UNIT ROOT TESTS <- ADVERTISING

```
Augmented Dickey-Fuller Test
alternative: stationary
 Type 1: no drift no trend
     lag   ADF p.value
[1,]   0 -2.50  0.0139
[2,]   1 -2.29  0.0227
[3,]   2 -2.00  0.0458
[4,]   3 -1.92  0.0541
[5,]   4 -1.60  0.1057
Type 2: with drift no trend
     lag   ADF p.value
[1,]   0 -5.85    0.01
[2,]   1 -5.62    0.01
[3,]   2 -4.97    0.01
[4,]   3 -5.08    0.01
[5,]   4 -3.95    0.01
Type 3: with drift and trend
     lag   ADF p.value
[1,]   0 -6.20    0.01
[2,]   1 -5.99    0.01
[3,]   2 -5.35    0.01
[4,]   3 -5.50    0.01
[5,]   4 -4.36    0.01
```

```
Phillips-Perron Unit Root Test
alternative: stationary

Type 1: no drift no trend
 lag Z rho p.value
   4 -7.82  0.0534
-----
 Type 2: with drift no trend
 lag Z_rho p.value
   4 -60.8    0.01
-----
 Type 3: with drift and trend
 lag Z_rho p.value
   4 -67.2    0.01
```

```
KPSS Unit Root Test
alternative: nonstationary

Type 1: no drift no trend
 lag stat p.value
   3  2.4  0.0182
-----
 Type 2: with drift no trend
 lag   stat p.value
   3 0.351  0.0983
-----
 Type 1: with drift and trend
 lag    stat p.value
   3 0.0644     0.1
```

- ADF & PP Tests: Both tests reject the unit root hypothesis across drift and trend specifications ($p \le 0.01$).
- KPSS Test: The trend specification further validates these findings, failing to reject and thus confirming stationarity ($p \ge 0.10$).
- LnAdvertising is stationary

RESULTS: UNIT ROOT TESTS <- PRICE

```
Augmented Dickey-Fuller Test
Type 1: no drift no trend
     lag    ADF p.value
[1,]   0 -0.776   0.401
[2,]   1 -0.748   0.411
[3,]   2 -0.677   0.437
[4,]   3 -0.568   0.476
[5,]   4 -0.417   0.524
Type 2: with drift no trend
     lag   ADF p.value
[1,]   0 -4.65    0.01
[2,]   1 -4.91    0.01
[3,]   2 -5.09    0.01
[4,]   3 -4.35    0.01
[5,]   4 -4.94    0.01
Type 3: with drift and trend
     lag   ADF p.value
[1,]   0 -5.64    0.01
[2,]   1 -6.08    0.01
[3,]   2 -6.45    0.01
[4,]   3 -5.79    0.01
[5,]   4 -6.47    0.01
```

```
Phillips-Perron Unit Root Test
alternative: stationary

Type 1: no drift no trend
 lag Z rho p.value
   4 -1.19   0.472
-----
 Type 2: with drift no trend
 lag Z_rho p.value
   4 -42.1    0.01
-----
 Type 3: with drift and trend
 lag Z_rho p.value
   4 -62.5    0.01
```

```
KPSS Unit Root Test
alternative: nonstationary

Type 1: no drift no trend
 lag  stat p.value
   3 0.354     0.1
-----
 Type 2: with drift no trend
 lag  stat p.value
   3 0.592  0.0234
-----
 Type 1: with drift and trend
 lag stat p.value
   3 0.02     0.1
```

- ADF & PP: Both tests reject the unit root ($p \leq 0.01$ with drift and trend).
- KPSS: Not required, as ADF/PP already rejected the unit root.
- LnPrice is stationary

RESULTS:UNIT ROOT TESTS <- COMPETITOR ADV

```
Augmented Dickey-Fuller Test
Type 1: no drift no trend
     lag   ADF p.value
[1,]   0 -2.53  0.0126
[2,]   1 -1.41  0.1728
[3,]   2 -1.35  0.1966
[4,]   3 -1.27  0.2241
[5,]   4 -1.26  0.2285
Type 2: with drift no trend
     lag   ADF p.value
[1,]   0 -8.23    0.01
[2,]   1 -4.47    0.01
[3,]   2 -4.37    0.01
[4,]   3 -4.17    0.01
[5,]   4 -4.36    0.01
Type 3: with drift and trend
     lag   ADF p.value
[1,]   0 -8.60    0.01
[2,]   1 -4.69    0.01
[3,]   2 -4.60    0.01
[4,]   3 -4.41    0.01
[5,]   4 -4.64    0.01
```

```
Phillips-Perron Unit Root Test
alternative: stationary

Type 1: no drift no trend
 lag Z_rho p.value
   4 -5.87  0.0958
-----
 Type 2: with drift no trend
 lag Z_rho p.value
   4  -117    0.01
-----
 Type 3: with drift and trend
 lag Z_rho p.value
   4  -128    0.01
```

```
KPSS Unit Root Test
alternative: nonstationary

Type 1: no drift no trend
 lag stat p.value
   3 4.53    0.01
-----
 Type 2: with drift no trend
 lag  stat p.value
   3 0.502  0.0412
-----
 Type 1: with drift and trend
 lag   stat p.value
   3 0.0759     0.1
```

- Both ADF and PP tests reject the unit root hypothesis.
- The KPSS test confirms this by failing to reject stationarity
- The variable LnCompAdvertising is trend-stationary

RESULTS:UNIT ROOT TESTS <- COMPETITOR PRICE

```
Augmented Dickey-Fuller Test
 Type 1: no drift no trend
       lag   ADF p.value
[1,]     0 -1.89  0.0586
[2,]     1 -1.93  0.0525
[3,]     2 -2.04  0.0421
[4,]     3 -2.32  0.0215
[5,]     4 -2.22  0.0268
Type 2: with drift no trend
       lag   ADF p.value
[1,]     0 -2.87  0.0529
[2,]     1 -2.04  0.3128
[3,]     2 -1.73  0.4317
[4,]     3 -1.24  0.6134
[5,]     4 -1.12  0.6535
Type 3: with drift and trend
       lag   ADF p.value
[1,]     0 -7.45  0.0100
[2,]     1 -5.32  0.0100
[3,]     2 -4.52  0.0100
[4,]     3 -3.25  0.0799
[5,]     4 -3.21  0.0868
```

```
Phillips-Perron Unit Root Test
alternative: stationary

Type 1: no drift no trend
 lag Z_rho p.value
   4 -2.46   0.367
-----
 Type 2: with drift no trend
 lag Z_rho p.value
   4 -7.47   0.314
-----
 Type 3: with drift and trend
 lag Z_rho p.value
   4 -86.3    0.01
```

```
KPSS Unit Root Test
alternative: nonstationary

Type 1: no drift no trend
 lag stat p.value
   3  1.5   0.067
-----
 Type 2: with drift no trend
 lag stat p.value
   3 1.53    0.01
-----
 Type 1: with drift and trend
 lag  stat p.value
   3 0.341    0.01
```

- PP test reject the unit root hypothesis in the trend specification p = 0.01.
- KPSS: Not required per methodological rules, as the unit root was already successfully rejected by PP.
- LnCompPrice is trend-stationary.

RESULTS: VAR MODEL

```
> endo_vars = ts_data[, c("LnSales", "LnAdvertising", "LnPrice", "LnCompAdvertising",
"LnCompPrice")]
> exo_vars = as.matrix(ts_data[, c("Qrtr1", "Qrtr2", "Qrtr3")])
> # Select optimal lags
> lag_selection <- VARselect(endo_vars, lag.max = 13, type = "both", exogen = exo_vars)
```

```
AIC(n)  HQ(n)  SC(n) FPE(n)
     2      1      1      2
```

A lag order of 1 (p = 1) was officially selected based on the Schwarz Criterion (SC).

RESULTS: VAR MODEL

```
> var_model <- VAR(endo_vars, p = 1, type = "both", exogen = exo_vars)
> summary(var_model, "LnSales")

VAR Estimation Results:
=========================
Endogenous variables: LnSales, LnAdvertising, LnPrice, LnCompAdvertising, LnCompPrice
Deterministic variables: both
Sample size: 206
Log Likelihood: -196.88
Roots of the characteristic polynomial:
0.7365 0.5681 0.4606 0.4606 0.2933
Call:
VAR(y = endo_vars, p = 1, type = "both", exogen = exo_vars)


Estimation results for equation LnSales:
========================================
LnSales = LnSales.l1 + LnAdvertising.l1 + LnPrice.l1 + LnCompAdvertising.l1 + LnCompPrice.l1 + const + trend + Qrtr1 + Qrtr2 + Qrtr3

                      Estimate  Std. Error t value Pr(>|t|)
LnSales.l1            0.2561987  0.0812757   3.152  0.00187 **
LnAdvertising.l1     -0.0065312  0.0035726  -1.828  0.06905 .
LnPrice.l1           -1.3924721  0.3462474  -4.022 8.24e-05 ***
LnCompAdvertising.l1 -0.0016947  0.0034993  -0.484  0.62871
LnCompPrice.l1       -0.2621387  0.3389575  -0.773  0.44024
const                 7.2423250  0.7972185   9.084  < 2e-16 ***
trend                 0.0007929  0.0006219   1.275  0.20380
Qrtr1                -0.0175370  0.0438785  -0.400  0.68983
Qrtr2                -0.0690197  0.0434683  -1.588  0.11394
Qrtr3                -0.0848416  0.0439839  -1.929  0.05518 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.1979 on 196 degrees of freedom
Multiple R-Squared: 0.4588,     Adjusted R-squared: 0.4339
F-statistic: 18.46 on 9 and 196 DF,  p-value: < 2.2e-16
```

RESULTS: VAR MODEL

Covariance matrix of residuals:

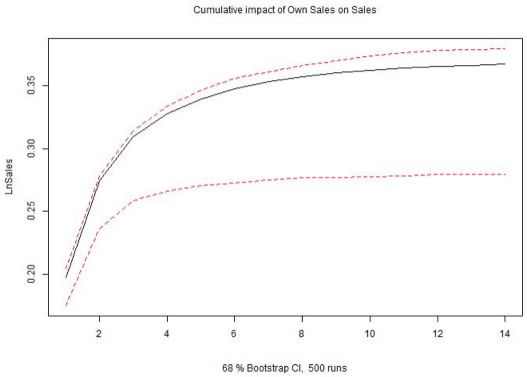
	LnSales	LnAdvertising	LnPrice	LnCompAdvertising	LnCompPrice
LnSales	3.917e-02	0.033326	-0.003922	0.0722163	2.213e-05
LnAdvertising	3.333e-02	9.425292	0.007550	0.5105587	-5.817e-03
LnPrice	-3.922e-03	0.007550	0.001479	-0.0061002	1.980e-04
LnCompAdvertising	7.222e-02	0.510559	-0.006100	13.4127264	3.837e-04
LnCompPrice	2.213e-05	-0.005817	0.000198	0.0003837	1.182e-03

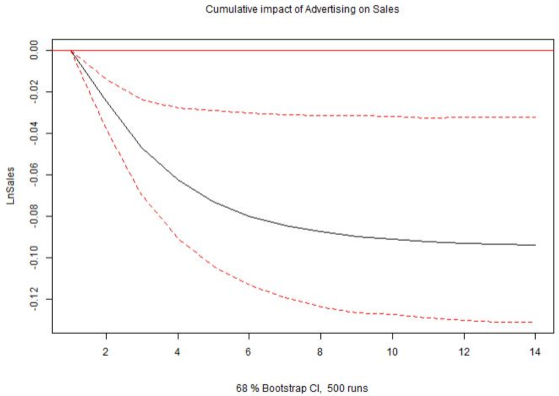
Correlation matrix of residuals:

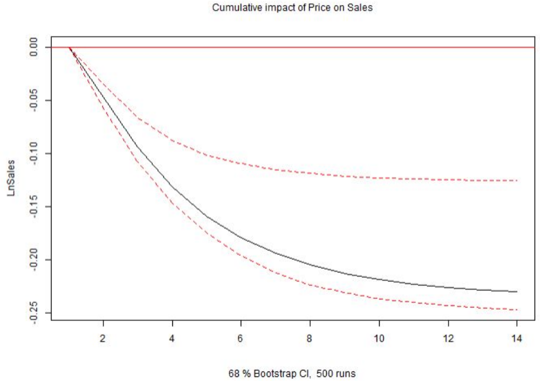
	LnSales	LnAdvertising	LnPrice	LnCompAdvertising	LnCompPrice
LnSales	1.000000	0.05485	-0.51524	0.099633	0.003252
LnAdvertising	0.054847	1.00000	0.06394	0.045409	-0.055105
LnPrice	-0.515240	0.06394	1.00000	-0.043307	0.149738
LnCompAdvertising	0.099633	0.04541	-0.04331	1.000000	0.003047
LnCompPrice	0.003252	-0.05510	0.14974	0.003047	1.000000

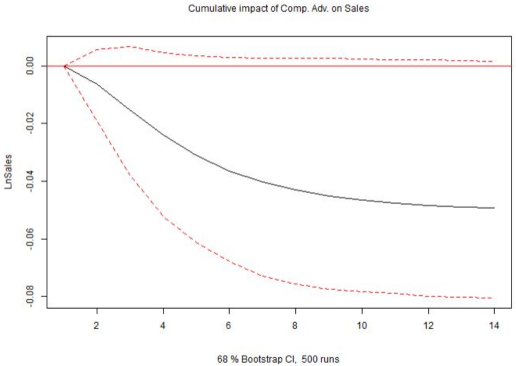
- The dynamic system is mathematically stable (max root = 0.74) and successfully explains ~46% of the total variance in weekly sales.
- Price is the Primary Driver: Own price exerts a massive, highly significant negative impact on sales ($p < 0.001$), with a short-term
- elasticity of -1.39. Consumers are price-sensitive.
- Past sales positively drive current sales ($p < 0.01$), proving that behavioral loyalty and consumer habits strongly protect our baseline volume.
- Own advertising shows a marginally significant, *negative* effect ($p < 0.10$). Current ad spend seems inefficient.
- Competitor pricing and advertising have absolutely no significant impact on our sales ($p > 0.40$).

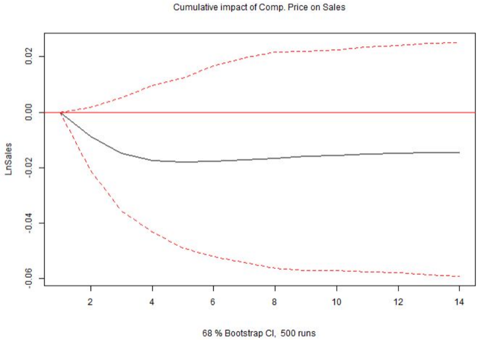
RESULTS: CUMULATIVE IRF'S









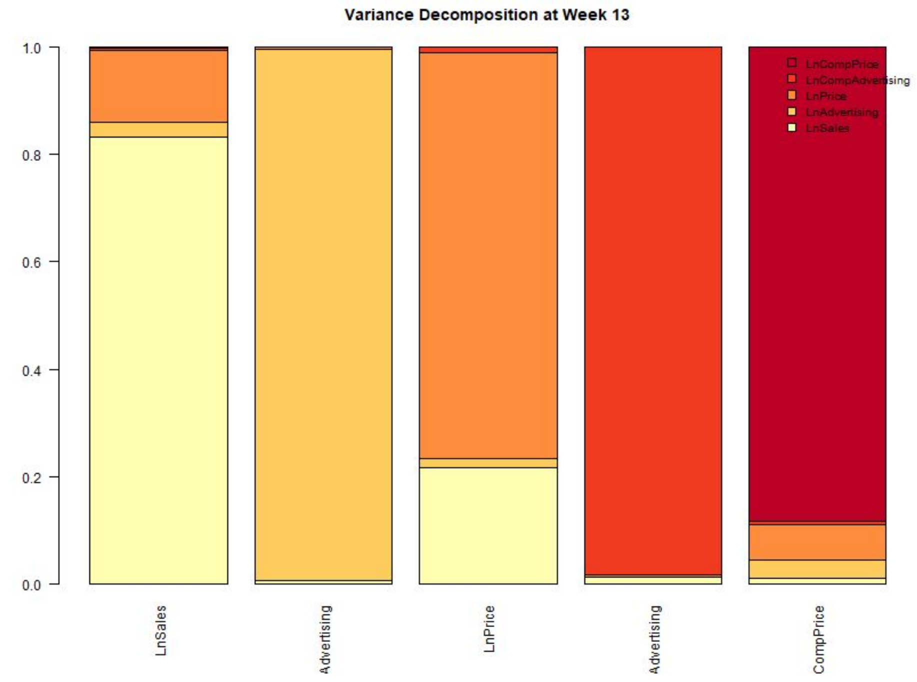


RESULTS: CUMULATIVE IRF'S

- Pricing Impact: A sudden price increase causes a severe, immediate drop in sales. The negative shock is strong but short-lived, absorbed by the market within 2 weeks.
- Sales Momentum : A positive sales shock generates significant short-term inertia, carrying extra volume into the immediate following week before stabilizing.
- Advertising: Changes in advertising spend do not currently yield a statistically significant volume uplift across the tested periods, highlighting a clear opportunity to review and optimize future media investments.
- Competitors: Shocks from competitor activities remain statistically insignificant across all periods, confirming our sales volume is independent of short-term competitive pressures.

RESULTS: FEVD

```
> lapply(fevd_sales, function(x) x[c(1, 6, 13), ])
$LnSales
       LnSales LnAdvertising   LnPrice LnCompAdvertising LnCompPrice
[1,] 1.0000000    0.00000000 0.0000000       0.000000000 0.000000000
[2,] 0.8397918    0.02706689 0.1260581       0.004916417 0.002166806
[3,] 0.8320350    0.02747180 0.1329036       0.005419256 0.002170377

$LnAdvertising
         LnSales LnAdvertising     LnPrice LnCompAdvertising  LnCompPrice
[1,] 0.003008219     0.9969918 0.000000000      0.0000000000 0.0000000000
[2,] 0.007803102     0.9874189 0.003580042      0.0007194135 0.0004785488
[3,] 0.007798219     0.9870260 0.003915922      0.0007445558 0.0005153032

$LnPrice
       LnSales LnAdvertising   LnPrice LnCompAdvertising LnCompPrice
[1,] 0.2654724   0.008526679 0.7260009       0.000000000 0.000000000
[2,] 0.2184842   0.017078037 0.7539167       0.009877803 0.000643235
[3,] 0.2163566   0.017607909 0.7546048       0.010669895 0.000760765

$LnCompAdvertising
         LnSales LnAdvertising      LnPrice LnCompAdvertising  LnCompPrice
[1,] 0.009926646   0.001600353 2.586471e-05         0.9884471 0.0000000000
[2,] 0.012562226   0.001591707 3.336128e-03         0.9822302 0.0002797873
[3,] 0.012564876   0.001613739 3.393505e-03         0.9821473 0.0002805390

$LnCompPrice
          LnSales LnAdvertising    LnPrice LnCompAdvertising LnCompPrice
[1,] 1.057331e-05   0.003065454 0.03374712      1.621571e-05   0.9631606
[2,] 1.061293e-02   0.030986795 0.05791999      7.942213e-03   0.8925381
[3,] 1.191150e-02   0.032277823 0.06592864      7.960214e-03   0.8819218
```



The main drivers of sales are primarily past sales and the pricing strategy. Advertising doesn’t provide any significant volume uplift. Competitor actions show no significant impact.

03 APPENDIX

REFERENCE LIST

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1. Gijsenberg, M. J. (2025). *Statistical Learning in Marketing: Marketing Dynamics, Part A*. Department of Marketing, Faculty of Economics and Business, University of Groningen.

2. Gijsenberg, M. J. (2025). *Statistical Learning in Marketing: Marketing Dynamics, Part B*. Department of Marketing, Faculty of Economics and Business, University of Groningen.

3. Shapiro, B. T., Hitsch, G. J. & Tuchman, A. E. (2021). TV Advertising Effectiveness and Profitability: Generalizable Results From 288 Brands. Econometrica, 89(4), 1855-1879.

4. Gauri, D. K., Ratchford, B., Pancras, J. & Talukdar, D. (2017). An Empirical Analysis of the Impact of Promotional Discounts on Store Performance. Journal of Retailing, 93(3), 283-303.

5. Steenkamp, J. B. E. & Slotegraaf, R. J. (2019). Retaliation and Business Resilience. Journal of Marketing, 83(4), 32-49.

6. Datta, H., Ailawadi, K. L. & van Heerde, H. J. (2017). How Well Does Consumer-Based Brand Equity Align with Sales-Based Brand Equity and Marketing Mix Response? Journal of Marketing, 81(3), 1-20.

7. Lamey, L., Deleersnyder, B., Steenkamp, J. B. E. & Dekimpe, M. G. (2018). New Product Success: The Role of Business Cycle Timing. Journal of Marketing Research, 55(1), 1-15.